Learning Latent Fields in Multiobject Tracking

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***Abstract*—Multiobject tracking using measurements from arrays of wireless sensors is a key task in applications for modern convenience and public safety. Many existing multiobject tracking approaches perform preprocessing steps and compress raw complex-valued signals into detections or intensity maps, discarding potentially relevant information about the objects and the propagation environment. We propose a Bayesian framework for multiobject tracking that relates object states to raw complex-valued signals via a physics-based superpositional model. Our statistical model combines a physics-based array manifold that represents the complex response of the sensors and a latent field, i.e., a family of context-dependent signal-power distributions that represents the effects of the objects and the propagation environment and is shared across objects and time. The latent field is parameterized by a conditional normalizing flow that is trained during runtime without labeled data by maximizing an evidence lower bound (ELBO) derived from the Bayesian model, coupling learning and inference in a sequential framework. We evaluate the method in simulation and in real-world radio and acoustic environments, with spatial scales ranging from meters to kilometers.**

***Index Terms*—Multiobject tracking, Bayesian framework, superpositional model, normalizing flow, wireless sensor arrays.**

I. Introduction

Multiobject tracking from received signals is a core problem in wireless sensing systems [1], [2]. Typically, objects and the shared propagation environment shape the received waveforms through diffraction, scattering, and reflection. As a result, the raw complex-valued signals received by an array of wireless sensors contain signal-strength variations, interference patterns, and background returns that reflect not only object locations but also the propagation environment. Fig. 1 illustrates this setting.

Conventional methods for multiobject tracking typically process data in two stages. A front-end detector based on beamforming, statistical parameter estimation, or learned regression compresses the high-dimensional raw complex-valued signals into geometric parameters (e.g., delays and angles) [3]–[5] or feature representations (e.g., class and shape) [6], [7]. A downstream tracking method then estimates object states from these detection results using a model of the object dynamics [8], [9]. A full complex-valued fast Fourier transform (FFT) changes the measurement coordinates while preserving the information in the input samples. Information loss can arise from magnitude conversion, selection or projection of the data, and reduction to point detections or other compressed features.

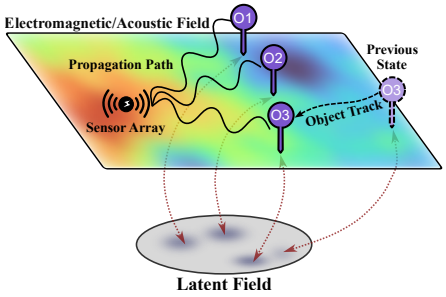


Fig. 1. Objects O1, O2, and O3 moving within a shared propagation environment generate raw complex-valued measurements at an array of wireless sensors. Our approach aims to learn a latent field of signal-power statistics that characterize object and propagation effects to improve multiobject tracking performance.

A natural way to reduce information loss is to avoid a two-stage processing approach and perform multiobject tracking using richer measurement representations, such as signal intensity maps. This strategy, known as track-before-detect (TkBD), has been explored using both model-based and learning-based approaches [10]–[12]. BP-TkBD [13], a TkBD method based on belief propagation (BP), operates directly on raw complex-valued sensor measurements. Model-based TkBD methods, including BP-TkBD, describe the effects of the propagation environment on the measurements using prescribed models. Modeling the propagation environment can be challenging, making the resulting tracking methods sensitive to model mismatch.

To learn propagation statistics during multiobject tracking, we build on the following observations:

1) Objects at nearby positions in a shared environment experience similar signal propagation conditions.
2) Multiobject tracking implicitly explores the propagation environment by observing objects at many locations.
3) A statistical surrogate of object and propagation effects on signal power can improve tracking performance.

Motivated by these observations, we combine a physics-based array manifold with learnable signal-power PDFs in a superpositional measurement model. In particular, we learn the statistical surrogate in observation 3 as a family of context-dependent signal-power PDFs that we refer to as the latent field. The context consists of the object's position and possibly sensing parameters and signal features. The parameters of the latent field are shared across objects and time. For a fixed context, the signal-power PDFs summarize the combined effects of propagation and object properties, such as source strength or reflectivity.

At each time step, particle-based BP [14], [15] approximates the posterior PDFs of object states, existence variables, signal powers, and noise variance. We use the inferred existence probabilities and signal-power PDFs to update a conditional

This work was supported by the National Science Foundation (NSF) under Award Nos. 2146261 and 2506477. Parts of this paper were presented at FUSION 2026, Trondheim, Norway, June 2026.

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normalizing flow [16] through an objective derived from the evidence lower bound (ELBO). The normalizing flow represents the latent field, which is thus continuously updated while objects are tracked. More details on latent field modeling are discussed in Section III-C.

II. Related Work and Contributions

Object detection and multiobject tracking from wireless signals have been studied extensively using both model-based and learning-based methods.

A. Object Detection

Various methods exist for extracting object location information from wireless signals. Model-based approaches include matched filtering [17], beamforming [18], subspace methods such as multiple signal classification (MUSIC) [19], and sparse Bayesian learning (SBL) [20], [21]. Constant false-alarm rate (CFAR) detectors [22] adapt detection thresholds to local clutter statistics. Array-processing methods such as beamforming, MUSIC, and SBL use physics-based manifolds to relate the sensor response to object location parameters. Some of these model-based approaches also employ a statistical noise model. Learning-based approaches instead establish mappings from signal-derived features to locations. Examples include fingerprinting methods in wireless communications [23], [24] and neural networks that detect objects in FFT-derived radar maps. RODNet [7] uses complex range–angle (RA) maps, and RAMP-CNN [6] combines complex RA maps with magnitude maps in the range–velocity (RV) and velocity–angle (VA) domains.

B. Multiobject Tracking

In detect-then-track multiobject tracking [25], object states are estimated from detections over time. Model-based approaches describe data association and state evolution probabilistically. Examples include joint probabilistic data association (JPDA) filters [26], methods based on random finite sets (RFSs) [8], [27], [28], and BP-based tracking methods [29]–[32]. Conventional JPDA assumes a known number of objects and requires separate track-management procedures or existence-aware extensions to handle object appearance and disappearance. RFS and BP formulations can explicitly represent uncertain object existence.

Learning-based architectures have also been developed for visual tracking. The approach in [33] jointly learns detection and across-frame track regression using a convolutional neural network (CNN). Graph neural networks learn associations between image-based detections [34]. The transformer-based MOTR [35] performs end-to-end visual tracking by representing each object with a learned feature vector that is updated using the image features of each new frame. These architectures are trained on labeled images or trajectories and provide precedents for learning-based tracking.

For tracking methods that operate on point detections, information omitted by the detector is unavailable for the Bayesian multiobject tracking method. Additional information loss may be the result of additional data reduction or inference approximations.

The alternative strategy, TkBD [10], [11], [13], [36], performs inference on lower-level measurements, but it can involve high computational complexity and can be sensitive to model mismatch. Several TkBD methods obtain tractable inference through specific modeling choices. Histogram probabilistic multihypothesis tracking (H-PMHT) [37], [38] models intensity-image cell energies as a histogram of synthetic point counts. Conventional implementations add procedures for track initialization and termination. The method in [39] derives tractable image-based updates by assuming that the regions influenced by different objects do not overlap. The Poisson/multi-Bernoulli approach in [40] supports scalable Bayesian TkBD inference using a measurement model in which each object contributes to one cell. Many TkBD methods have been evaluated mainly in simulation, so their performance in complex real-world propagation environments is less explored. BP-TkBD [13] uses a superpositional model of raw complex-valued sensor measurements and jointly infers object kinematics, existence, signal powers, and noise variances through particle-based BP with Gaussian moment matching. Its object-specific signal-power priors are propagated using prescribed state-transition models. We retain the superpositional model with conditionally Gaussian amplitudes but replace these prescribed state-transition models by the latent field. The latent field is learned during runtime and shares information about object and propagation effects across the tracked objects.

C. Contributions

Learning statistical representations of signal propagation has precedents in wireless localization. Gaussian-process models learn location-dependent signal-strength likelihoods from calibration data [41]. WiFi simultaneous localization and mapping (WiFi-SLAM) jointly estimates a device trajectory and signal-strength maps from unlabeled measurement sequences, while using kernel parameters learned beforehand from labeled data [42]. Multisnapshot SBL also estimates source-power and noise-variance hyperparameters from complex array measurements through evidence maximization [21]. The proposed framework combines a physics-based superpositional model of raw complex-valued sensor data with Bayesian multiobject tracking and a shared, context-dependent latent field represented by a conditional normalizing flow. We learn this field during runtime through an ELBO-derived objective without labeled data or offline pretraining. To the best of our knowledge, this combination has not been proposed in the multiobject tracking literature.

Whereas existing formulations generally treat the signal amplitude or power as part of the object states or features, in our formulation each object contributes information to the latent field, which can also be evaluated at contexts not encountered during tracking.

Our main contributions are summarized as follows:

1) We formulate a superpositional model of raw complex-valued array signals with a physics-based array manifold and the shared latent field.

2) We combine BP tracking with learning the latent field, represented by a conditional normalizing flow, during runtime without labeled data or offline pretraining.
3) We develop an efficient likelihood evaluation that exploits low-rank and Kronecker structure, with a complexity that scales linearly in the measurement dimension.
4) We demonstrate improved tracking performance on real radar and acoustic data and quantify the effect of individual model components through ablation studies.

This paper advances over the preliminary account of our method provided in the conference publication [43] by (i) presenting a detailed derivation of the learning objective that accounts for the uncertain existence of objects, (ii) developing an efficient likelihood evaluation that exploits low-rank and Kronecker structure, together with a complexity and runtime analysis, (iii) demonstrating performance advantages using real radar and acoustic data, (iv) comparing the learned latent field with physics-based predictions of the KRAKEN normal-mode program, and (v) assessing the design of the signal-power prior in a controlled ablation study.

The remainder of this paper is organized as follows. Section III formulates the multiobject tracking problem and introduces the physics-based superpositional model and the statistical model of the latent field. Section IV presents Bayesian inference, the learning objective with its parameter updates, and the conditional normalizing flow. Section V evaluates the proposed method in a simulated radar scenario and using real radar and acoustic data, including a controlled acoustic simulation. Section VI concludes the paper. Due to space limitations, detailed derivations of the learning objective and of the BP-based inference, the efficient likelihood evaluation and its complexity analysis, as well as additional numerical results have been omitted and can be found in the supplementary material [44].

Notation: Scalars are denoted by italic letters, vectors by bold lowercase letters, e.g., $\mathbf{x}$ and $\boldsymbol{\gamma}$, and matrices and tensors by bold uppercase letters, e.g., $\boldsymbol{\Sigma}_k$ and $\mathbf{Z}_k$. The sets of real, positive real, and complex numbers are denoted by $\mathbb{R}$, $\mathbb{R}_{>0}$, and $\mathbb{C}$, respectively. The superscript $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose, $\mathbf{I}_C$ the $C \times C$ identity matrix, and $\mathrm{vec}(\cdot)$ the vectorization of a tensor. The sequence $\{\mathbf{z}_k\}_{k=1}^K$ is written as $\mathbf{z}_{1:K}$. Probability density functions (PDFs) and probability mass functions are denoted by $f(\cdot)$, approximate posterior PDFs by $q(\cdot)$, and a subscript $\boldsymbol{\theta}$ indicates dependence on the learnable parameters. The zero-mean proper complex Gaussian PDF with covariance matrix $\boldsymbol{\Sigma}$ is denoted by $\mathcal{CN}(\mathbf{0}, \boldsymbol{\Sigma})$, and its PDF evaluated at $\mathbf{z}$ by $\mathcal{CN}(\mathbf{z}; \mathbf{0}, \boldsymbol{\Sigma})$. The standard Gaussian PDF is denoted by $\mathcal{N}(0, 1)$. Finally, $\mathbb{E}_q[\cdot]$ denotes the expectation with respect to $q(\cdot)$, and $\mathrm{KL}(\cdot \,\|\, \cdot)$ denotes the Kullback–Leibler divergence.

III. Problem Formulation and Statistical Model

We formulate the multiobject tracking problem, specify the superpositional measurement model, and introduce the statistical model of the latent field.

A. Problem Formulation

a) Scenario: We consider a sequence of sensor measurements received over discrete time steps $k = 1, \ldots, K$. Objects may appear and disappear over time. We represent the unknown and time-varying number of objects using N_k potential objects. Potential object $n \in \{1, \ldots, N_k\}$ has a kinematic state $\mathbf{x}_{k,n} \in \mathbb{R}^d$ (e.g., position and velocity) and a binary existence variable $r_{k,n} \in \{0, 1\}$. The number of existing objects is $L_k = \sum_{n=1}^{N_k} r_{k,n}$. In contrast to the known number N_k of potential objects, L_k is an unknown random variable. At each time step k, the sensor array produces a complex-valued measurement tensor $\mathbf{Z}_k \in \mathbb{C}^{C_1 \times \cdots \times C_m}$. We denote the vectorized representation of the measurement tensor by $\mathbf{z}_k = \mathrm{vec}(\mathbf{Z}_k) \in \mathbb{C}^C$, where $C = \prod_{j=1}^m C_j$.

b) Goal: At each time step k, our goal is to infer the number of objects and their states from the raw complex-valued measurements $\mathbf{z}_{1:k}$ received up to that time. Because sequential estimation of object states implicitly explores the propagation environment, it provides the information needed to learn the latent field, which in turn can improve subsequent inference.

B. Physics-Based Superpositional Model

We review the single-dictionary, single-snapshot case of the measurement model in [13, Sec. II-A], omitting the dictionary and snapshot indices. Here, a dictionary specifies the state-dependent array response, a snapshot is one of several measurement vectors acquired at the same time step, and vectorization collects the samples within one measurement block. The extension to multiple dictionaries and snapshots follows the conditional block-likelihood factorization described therein.

At each time step k, each existing object contributes a complex array response determined by the physics-based array manifold, while propagation effects are captured through a stochastic complex amplitude. The raw signal $\mathbf{z}_k$ is modeled as a coherent superposition of the contributions from existing objects, represented by the N_k potential-object terms, plus noise, i.e.,

$$\mathbf{z}_k \;=\; \sum_{n=1}^{N_k} r_{k,n}\, \rho_{k,n}\, \mathbf{a}(\mathbf{x}_{k,n}) \;+\; \mathbf{w}_k. \tag{1}$$

Here, the physics-based array manifold $\mathbf{a}(\cdot)$ maps an object state $\mathbf{x}_{k,n}$ to the corresponding complex array response $\mathbf{a}(\mathbf{x}_{k,n}) \in \mathbb{C}^C$. The noise $\mathbf{w}_k$ is zero-mean proper complex Gaussian with covariance $\eta_k \mathbf{I}_C$, with $\eta_k > 0$ and unknown. The diagonal covariance models a white effective background across measurement entries.

The complex amplitude $\rho_{k,n} \in \mathbb{C}$ captures the unknown gain and phase of object n, representing aggregate gain effects compatible with the array response prescribed by the array manifold. It is modeled as a zero-mean proper complex Gaussian random variable, i.e.,

$$\rho_{k,n} \mid \gamma_{k,n} \sim \mathcal{CN}(0, \gamma_{k,n}) \tag{2}$$

where the positive random variable $\gamma_{k,n}$ is the conditional mean squared magnitude of the complex amplitude, which we refer to as the signal power. For an existing object, $\gamma_{k,n}$ is

TABLE I
MEMORY REQUIREMENT FOR A SINGLE LIKELIHOOD EVALUATION IN A TYPICAL RADAR SCENARIO WITH $C = 128 \times 256 \times 16 = 524{,}288$ COMPLEX VALUES STORED WITH 64 BITS EACH.

Item	Description	Memory Cost
Input measurement $\mathbf{z}_k$	C complex values	≈ 4 MiB
Direct computation (3)	Full covariance matrix	≈ 2.0 TiB
Efficient computation [44]	Decomposed components	≈ 3.1 KiB

given by the latent field evaluated at the object's context, as detailed in Section III-C. For an indexed quantity $u_{k,n}$, we write $u_{k,1:N_k} = (u_{k,1}, \dots, u_{k,N_k})$ for its ordered collection. In particular, $\mathbf{x}_k = \mathbf{x}_{k,1:N_k}$, $\mathbf{r}_k = r_{k,1:N_k}$, and $\boldsymbol{\gamma}_k = \gamma_{k,1:N_k}$ collect the kinematic states, existence variables, and signal powers. Conditional on these variables and η_k, the amplitudes $\rho_{k,n}$ are mutually independent with the PDFs in (2), and the noise is independent of all amplitudes with the PDF stated above. Marginalizing the amplitudes therefore yields the joint likelihood of the measurement entries,

$$f(\mathbf{z}_k \mid \mathbf{r}_k, \mathbf{x}_k, \boldsymbol{\gamma}_k, \eta_k) = \mathcal{CN}\big(\mathbf{z}_k; \mathbf{0}, \boldsymbol{\Sigma}_k\big) \tag{3}$$

$$\boldsymbol{\Sigma}_k = \sum_{n=1}^{N_k} r_{k,n}\gamma_{k,n}\mathbf{a}(\mathbf{x}_{k,n})\mathbf{a}(\mathbf{x}_{k,n})^{\mathrm{H}} + \eta_k \mathbf{I}_C. \tag{4}$$

The signal-dependent part of $\boldsymbol{\Sigma}_k$ has rank at most $\min(C, L_k)$, while $\eta_k > 0$ makes the full covariance positive definite. The generally nonzero off-diagonal entries retain correlations among measurements within this block. However, directly evaluating this Gaussian likelihood is generally intractable in the sensing regimes considered here, since the measurement dimension C can reach several hundred thousand. For example, a typical radar setup with 128 fast-time samples, 256 chirps, and 16 antennas yields $C = 524{,}288$. As illustrated in Table I, storing the full covariance matrix for a single likelihood evaluation then requires approximately 2 TiB, which is infeasible on standard hardware. We therefore exploit the low-rank signal contribution together with the diagonal noise term and, when available, a Kronecker factorization of the array manifold. This makes likelihood evaluation tractable for typical problem sizes without forming or inverting a full $C \times C$ covariance matrix.

C. Statistical Model of the Latent Field

We formalize the latent field as a positive-valued stochastic field indexed by context. The context of an object comprises its position and possibly known sensing parameters and signal features. Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\mathcal{C}$ denote the context space. For learnable parameters $\boldsymbol{\theta}$, the latent field is a collection of positive random variables

$$\Gamma_{\boldsymbol{\theta}} = \big\{\Gamma_{\boldsymbol{\theta}}(\,\cdot\,, \mathbf{c})\big\}_{\mathbf{c}\in\mathcal{C}} \qquad \Gamma_{\boldsymbol{\theta}}(\,\cdot\,, \mathbf{c}) : \Omega \longrightarrow \mathbb{R}_{>0}. \tag{5}$$

For any fixed context $\mathbf{c} \in \mathcal{C}$, $\Gamma_{\boldsymbol{\theta}}(\,\cdot\,, \mathbf{c})$ is a positive random variable. Similarly, for any fixed outcome $\omega \in \Omega$, the function $\Gamma_{\boldsymbol{\theta}}(\omega, \,\cdot\,) : \mathcal{C} \to \mathbb{R}_{>0}$ is one realization of the latent field over the context space. For each fixed context $\mathbf{c}$, the random variable $\Gamma_{\boldsymbol{\theta}}(\,\cdot\,, \mathbf{c})$ has the PDF $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})$. For each pair (k, n) of time step and potential object, we introduce an independent copy $\Gamma_{\boldsymbol{\theta}}^{(k,n)}$ of this field. The copies share the deterministic parameters $\boldsymbol{\theta}$ and the PDF $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})$ at each context. If object n exists at time k, its signal power is obtained by evaluating its copy at the context $\mathbf{c}_{k,n}$, i.e.,

$$\gamma_{k,n} = \Gamma_{\boldsymbol{\theta}}^{(k,n)}(\,\cdot\,, \mathbf{c}_{k,n}). \tag{6}$$

Thus, through the PDFs $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})$, $\mathbf{c} \in \mathcal{C}$, the latent field defines a family of context-dependent signal-power PDFs. Parameters shared across objects and time steps allow observations from multiple object tracks to update the latent field jointly.

By (2), the signal power satisfies

$$\mathbb{E}\big[|\rho_{k,n}|^2 \mid \gamma_{k,n}\big] = \gamma_{k,n}. \tag{7}$$

The latent field therefore models the statistics of the combined object and propagation effects on the signal power relative to the array response $\mathbf{a}(\mathbf{x}_{k,n})$, such as source strength, reflectivity, attenuation, and multipath-induced gain variations.

The context $\mathbf{c}_{k,n}$ is treated as a fixed input during each inference and learning step. In our numerical studies in Section V, the context consists of a positional encoding of the object position estimated from the preceding posterior, together with known sensing parameters and signal features. In our implementation, the context-dependent PDFs are represented by the conditional normalizing flow discussed in Section IV-C.

Since $\gamma_{k,n}$ affects the measurement covariance in (4) only if $r_{k,n} = 1$, we consider its PDF conditioned on $r_{k,n} = 1$, denoted by

$$f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n} = 1; \mathbf{c}_{k,n}). \tag{8}$$

We combine the random variables $r_{k,n}$ and $\gamma_{k,n}$ as $\boldsymbol{\phi}_{k,n} = (r_{k,n}, \gamma_{k,n})$ to jointly represent object existence and signal power. The PDF of $\boldsymbol{\phi}_{k,n}$ is given by

$$f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n}) = \begin{cases} f(r_{k,n} = 1)\, f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n} = 1; \mathbf{c}_{k,n}) & r_{k,n} = 1 \\ f(r_{k,n} = 0)\, f_{\mathrm{D}}(\gamma_{k,n}) & r_{k,n} = 0 \end{cases} \tag{9}$$

where $f_{\mathrm{D}}(\gamma_{k,n})$ is a dummy PDF. Since the likelihood (3) does not depend on $\gamma_{k,n}$ if $r_{k,n} = 0$, any normalized PDF that does not depend on $\boldsymbol{\theta}$ can be chosen for $f_{\mathrm{D}}(\gamma_{k,n})$ without affecting existence probabilities and state estimates [13] or the learning objective.

The existence variable $r_{k,n}$ is modeled as independent of the field copies. It only determines whether the corresponding copy models the signal power of object n at time k. If $r_{k,n} = 1$, then $\gamma_{k,n}$ is given by (6). If $r_{k,n} = 0$, the object contributes no power and $\gamma_{k,n}$ is distributed according to the dummy PDF $f_{\mathrm{D}}(\gamma_{k,n})$. Since $\mathbf{c}_{k,n}$ is fixed and $r_{k,n}$ is independent of the field copies, the conditional PDF (8) can be written as

$$f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n} = 1; \mathbf{c}_{k,n}) = f_{\boldsymbol{\theta}}(\gamma_{k,n}; \mathbf{c}_{k,n}). \tag{10}$$

We assume that, for fixed $\boldsymbol{\theta}$, the values of each field copy at distinct contexts are independent. The normalizing flow in Section IV-C provides a construction of this field and its independent copies. Hence, for any finite set of distinct contexts $\{\mathbf{c}_j\}_{j=1}^{J}$,

$$f_{\boldsymbol{\theta}}(\gamma_1, \dots, \gamma_J; \mathbf{c}_1, \dots, \mathbf{c}_J) = \prod_{j=1}^{J} f_{\boldsymbol{\theta}}(\gamma_j; \mathbf{c}_j). \tag{11}$$

Hence, spatial structure, such as similar signal-power PDFs at nearby positions, is captured by the dependence of $f_{\boldsymbol{\theta}}(\gamma;\mathbf{c})$ on the context $\mathbf{c}$ through the shared parameters $\boldsymbol{\theta}$. By (10) and (11), learning the latent field is equivalent to learning the conditional PDF (8) for all contexts. The parameters $\boldsymbol{\theta}$ are updated during runtime as discussed in Section IV-B.

IV. Bayesian Inference and Latent-Field Learning

We use the model in Section III to infer object states and learn the latent field during runtime. We first describe Bayesian inference, then derive the learning objective and parameter update, and finally specify the conditional normalizing flow.

A. Bayesian Inference

Let $\mathbf{h}_k = \big(\mathbf{x}_k, \boldsymbol{\phi}_k, \eta_k\big)$, with $\boldsymbol{\phi}_k = \boldsymbol{\phi}_{k,1:N_k}$, collect all random variables at time k. The joint prior PDF is denoted by $f_{\boldsymbol{\theta}}(\mathbf{h}_k)$. Note that we adopt a sequential filtering perspective, where the prior is the result of prediction and update steps at previous time steps. That is, $f_{\boldsymbol{\theta}}(\mathbf{h}_k)$ is a shorthand for the predicted PDF $f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_{1:k-1})$; for notational simplicity, we omit the conditioning on $\mathbf{z}_{1:k-1}$. The posterior PDF at time step k is given by Bayes' rule, i.e.,

$$f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k) = \frac{f(\mathbf{z}_k \mid \mathbf{h}_k) f_{\boldsymbol{\theta}}(\mathbf{h}_k)}{f_{\boldsymbol{\theta}}(\mathbf{z}_k)}. \tag{12}$$

Here, $f(\mathbf{z}_k \mid \mathbf{h}_k) = \mathcal{CN}\big(\mathbf{z}_k; \mathbf{0}, \boldsymbol{\Sigma}_k\big)$ is the likelihood function. The denominator $f_{\boldsymbol{\theta}}(\mathbf{z}_k)$ is the evidence, given by $f_{\boldsymbol{\theta}}(\mathbf{z}_k) = \int f(\mathbf{z}_k \mid \mathbf{h}_k) f_{\boldsymbol{\theta}}(\mathbf{h}_k) \mathrm{d}\mathbf{h}_k$.

As is common, we model the kinematic states and the noise variance as first-order Markov processes with state-transition PDFs $f(\mathbf{x}_{k,n} \mid \mathbf{x}_{k-1,n})$ and $f(\eta_k \mid \eta_{k-1})$. Following [13], we distinguish legacy potential objects $n \in \{1, \ldots, N_{k-1}\}$, which were already considered at time step $k-1$, from new potential objects $n \in \{N_{k-1}+1, \ldots, N_k\}$, which account for newly appearing objects. The predicted PDFs of the kinematic states of legacy potential objects and of the noise variance are given by

$$f(\mathbf{x}_{k,n}) = \int f(\mathbf{x}_{k,n} \mid \mathbf{x}_{k-1,n}) q(\mathbf{x}_{k-1,n}) \mathrm{d}\mathbf{x}_{k-1,n} \tag{13}$$

$$f(\eta_k) = \int f(\eta_k \mid \eta_{k-1}) q(\eta_{k-1}) \mathrm{d}\eta_{k-1}. \tag{14}$$

The prediction step is implemented using conventional particle-filtering operations [45]. As in [13], $f(\eta_k \mid \eta_{k-1})$ is a Gamma PDF with mean η_{k-1}. The existence variable of a legacy potential object evolves according to $f(r_{k,n} = 1 \mid r_{k-1,n} = 1) = p_{\mathrm{s}}$ and $f(r_{k,n} = 1 \mid r_{k-1,n} = 0) = 0$, where p_{s} is the probability of object survival. Thus, its predicted existence probability in (9) is $f(r_{k,n} = 1) = p_{\mathrm{s}}\, q(r_{k-1,n} = 1)$. For a new potential object, $f(r_{k,n} = 1) = p_{\mathrm{B},n}$ is the birth probability, and $f(\mathbf{x}_{k,n}) = f_{\mathrm{B},n}(\mathbf{x}_{k,n})$ is the birth PDF of its kinematic state. The birth PDFs used in our experiments are described in Section V. For both legacy and new potential objects, the prior PDF of the signal power is provided by the latent field as described in Section III-C.

Since the prediction step operates on the marginal posterior PDFs of time step $k-1$, the joint prior PDF at time step k is represented by the product of the predicted PDFs [13], i.e.,

$$f_{\boldsymbol{\theta}}(\mathbf{h}_k) = f(\eta_k) \prod_{n=1}^{N_k} f(\mathbf{x}_{k,n})\, f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n}) \tag{15}$$

where $f(\eta_k)$ is given by (14), $f(\mathbf{x}_{k,n})$ is given by (13) for legacy potential objects and by $f_{\mathrm{B},n}(\mathbf{x}_{k,n})$ for new potential objects, and $f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})$ is given by (9) with the predicted existence probability $f(r_{k,n} = 1)$ discussed above. The parameters $\boldsymbol{\theta}$ enter (15) only through the conditional PDFs $f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n} = 1; \mathbf{c}_{k,n})$ in (9).

Direct computation of (12) is generally intractable. We therefore approximate the posterior PDF $f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k)$ by a PDF $q(\mathbf{h}_k)$. This PDF serves as a formal auxiliary density; BP computes only its marginal posterior PDFs, which are represented by weighted particles. The resulting marginal posterior PDFs are obtained by marginalizing $q(\mathbf{h}_k)$, i.e.,

$$q(\mathbf{x}_{k,n}) = \int q(\mathbf{h}_k)\, \mathrm{d}\mathbf{h}_k^{\sim \mathbf{x}_{k,n}} \tag{16}$$

$$q(\boldsymbol{\phi}_{k,n}) = \int q(\mathbf{h}_k)\, \mathrm{d}\mathbf{h}_k^{\sim \boldsymbol{\phi}_{k,n}} \tag{17}$$

$$q(\eta_k) = \int q(\mathbf{h}_k)\, \mathrm{d}\mathbf{h}_k^{\sim \eta_k}. \tag{18}$$

Here, $\mathbf{h}_k^{\sim u}$ denotes all components of $\mathbf{h}_k$ except u, and integration with respect to the binary existence variables in $\mathbf{h}_k$ is understood as summation over their values. In BP, these marginal posterior PDFs are approximated by beliefs of the form

$$q(\mathbf{x}_{k,n}) \propto f(\mathbf{x}_{k,n})\, \ell_{\mathbf{x},n}(\mathbf{x}_{k,n})$$

$$q(\boldsymbol{\phi}_{k,n}) \propto f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\, \ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_{k,n})$$

$$q(\eta_k) \propto f(\eta_k)\, \ell_{\eta}(\eta_k)$$

where $\ell_{\mathbf{x},n}(\mathbf{x}_{k,n})$, $\ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_{k,n})$, and $\ell_{\eta}(\eta_k)$ denote the marginal likelihood terms associated with the kinematic state of object n, its existence and signal power, and the noise variance at time k, respectively. These marginal likelihoods are computed approximately using particle-based BP [13]. In particular, each marginal likelihood is approximated by a zero-mean complex Gaussian PDF via second-order moment matching, and the expected outer product $\mathbb{E}[\mathbf{a}(\mathbf{x}_{k,n})\mathbf{a}(\mathbf{x}_{k,n})^{\mathrm{H}}]$ is approximated by the rank-one matrix $\mathbb{E}[\mathbf{a}(\mathbf{x}_{k,n})]\,\mathbb{E}[\mathbf{a}(\mathbf{x}_{k,n})]^{\mathrm{H}}$ [44, Secs. II-C and III-A]; for Kronecker-structured array responses, the mean array response is further approximated by the Kronecker product of the means of its factors [44, Sec. III-D]. As in [13], potential object n is declared to exist at time step k if $q(r_{k,n} = 1) > T_{\mathrm{dec}}$, and potential objects with $q(r_{k,n} = 1) < T_{\mathrm{pru}}$ are pruned before the next prediction step. The object state estimates of declared potential objects are obtained from the marginal posterior PDFs $q(\mathbf{x}_{k,n})$ using the minimum mean-square error (MMSE) estimator.

B. Learning Objective and Parameter Updates

We derive the learning objective from the standard ELBO. At time step k, for any fixed PDF $q(\mathbf{h}_k)$, the log-evidence

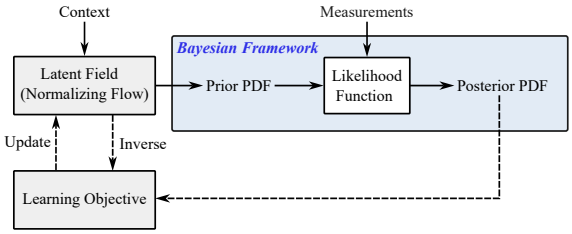


Fig. 2. The proposed inference and learning framework at one time step.

satisfies

$$\log f_{\boldsymbol{\theta}}(\mathbf{z}_k) = \mathcal{L}_k(\boldsymbol{\theta}) + \mathrm{KL}(q(\mathbf{h}_k) \,\|\, f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k)) \tag{19}$$

where

$$\mathcal{L}_k(\boldsymbol{\theta}) = \mathbb{E}_{q(\mathbf{h}_k)}\left[\log f(\mathbf{z}_k \mid \mathbf{h}_k) + \log f_{\boldsymbol{\theta}}(\mathbf{h}_k) - \log q(\mathbf{h}_k)\right].$$

Unlike conventional variational Bayes methods that restrict $q(\mathbf{h}_k)$ to a predefined parametric family, we do not specify $q(\mathbf{h}_k)$ explicitly; the learning objective below requires only its marginal posterior PDFs, which are represented by weighted particles provided by BP [44]. During the update of $\boldsymbol{\theta}$ at time step k, the predicted PDFs in (15), the contexts, and $q(\mathbf{h}_k)$ are held fixed, i.e., we do not differentiate through the computations of preceding time steps. By (15), the learnable parameters $\boldsymbol{\theta}$ appear only in the conditional PDF (8), which equals the signal-power PDF $f_{\boldsymbol{\theta}}(\gamma_{k,n}; \mathbf{c}_{k,n})$ of the latent field by (10). Hence, by (9), the part of $\log f_{\boldsymbol{\theta}}(\mathbf{h}_k)$ that depends on $\boldsymbol{\theta}$ is $\sum_{n=1}^{N_k} r_{k,n} \log f_{\boldsymbol{\theta}}(\gamma_{k,n}; \mathbf{c}_{k,n})$. Since the likelihood $f(\mathbf{z}_k \mid \mathbf{h}_k)$ and $q(\mathbf{h}_k)$ do not depend on $\boldsymbol{\theta}$, and the expectation of $r_{k,n} \log f_{\boldsymbol{\theta}}(\gamma_{k,n}; \mathbf{c}_{k,n})$ with respect to $q(\mathbf{h}_k)$ depends only on the marginal posterior $q(\boldsymbol{\phi}_{k,n})$, maximizing the ELBO with respect to $\boldsymbol{\theta}$ is equivalent to maximizing

$$\begin{aligned}\mathcal{J}_k(\boldsymbol{\theta}) = &\sum_{n=1}^{N_k} q(r_{k,n} = 1) \\ &\times \mathbb{E}_{q(\gamma_{k,n} \mid r_{k,n}=1)}\left[\log f_{\boldsymbol{\theta}}\big(\gamma_{k,n}; \mathbf{c}_{k,n}\big)\right]\end{aligned} \tag{20}$$

where $q(r_{k,n} = 1)$ and $q(\gamma_{k,n} \mid r_{k,n} = 1)$ are obtained from the marginal posterior $q(\boldsymbol{\phi}_{k,n})$. At time step k, the accumulated objective is given by

$$\mathcal{J}_{1:k}(\boldsymbol{\theta}) = \sum_{k'=1}^{k} \mathcal{J}_{k'}(\boldsymbol{\theta}) \tag{21}$$

which is evaluated across time and objects. The posterior quantities of preceding time steps, i.e., the existence probabilities $q(r_{k',n} = 1)$, the particles representing $q(\gamma_{k',n} \mid r_{k',n} = 1)$, and the contexts $\mathbf{c}_{k',n}$ with $k' < k$, are stored and held fixed; they are not recomputed with the updated parameters. At each time step k, $\boldsymbol{\theta}$ is initialized with its value from time step $k-1$ and updated by a few gradient-based optimization steps that increase $\mathcal{J}_{1:k}(\boldsymbol{\theta})$ (see Section V). This procedure can be interpreted as an approximate, generalized variational expectation-maximization algorithm [46], where the Bayesian inference step provides a particle-based approximate E-step and the normalizing flow update acts as the variational M-step. Fig. 2 summarizes the resulting inference/learning procedure at a single time step.

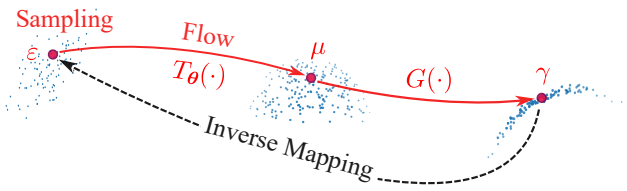


Fig. 3. Computational steps of the normalizing flow.

C. Conditional Normalizing Flow

As shown in Section III-C, learning the latent field amounts to learning the conditional PDFs $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})$, $\mathbf{c} \in \mathcal{C}$, which we parameterize by a conditional normalizing flow [16]. Since $\gamma_{k,n} > 0$, we first represent $\gamma_{k,n}$ by an unconstrained variable $\mu_{k,n}$, i.e.,

$$\gamma_{k,n} = G(\mu_{k,n}) \qquad \mu_{k,n} \in \mathbb{R} \tag{22}$$

where $G(\cdot) : \mathbb{R} \to \mathbb{R}_{>0}$ is a smooth bijection. For notational simplicity, we suppress the indices (k, n) in the following derivation. We then model the conditional PDF $f_{\boldsymbol{\theta}}(\mu; \mathbf{c})$ using a conditional normalizing flow. Let $f_0(\varepsilon)$ denote a base PDF on $\mathbb{R}$ (e.g., $\varepsilon \sim \mathcal{N}(0, 1)$). For a given context $\mathbf{c}$, we define a sequence of Q invertible, monotone transformations, i.e.,

$$y_0 = \varepsilon \qquad y_\ell = F_{\theta_\ell}(y_{\ell-1}; \mathbf{c}) \quad \ell = 1, \ldots, Q$$

where each $F_{\theta_\ell}(\cdot; \mathbf{c}) : \mathbb{R} \to \mathbb{R}$ is a differentiable bijection parameterized by θ_ℓ and conditioned on $\mathbf{c}$. In particular, $F_{\theta_\ell}(\cdot; \mathbf{c})$ is a nonlinear, strictly monotone function whose coefficients are computed from the context $\mathbf{c}$ by an encoder, i.e., a neural network with parameters θ_ℓ, such as a multilayer perceptron (MLP) or a CNN. Thus, the context determines the shape of each transformation, whereas the parameters $\boldsymbol{\theta}$ are shared across all contexts, objects, and time steps. The output of this sequence is given by

$$\mu = y_Q = T_{\boldsymbol{\theta}}(\varepsilon; \mathbf{c})$$

where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_Q)$ collects the parameters of all transformations. By the change-of-variables formula [47], the log-PDF of γ given $\mathbf{c}$ can be expressed as

$$\begin{aligned}\log f_{\boldsymbol{\theta}}(\gamma; \mathbf{c}) &= \log f_0(\varepsilon) - \log \left| \frac{\mathrm{d}G(T_{\boldsymbol{\theta}}(\varepsilon; \mathbf{c}))}{\mathrm{d}\varepsilon} \right| \\ &= \log f_0(\varepsilon) - \sum_{\ell=1}^{Q} \log \left| \frac{\partial F_{\theta_\ell}(y_{\ell-1}; \mathbf{c})}{\partial y_{\ell-1}} \right| \\ &\quad - \log \left| \frac{\mathrm{d}G(\mu)}{\mathrm{d}\mu} \right|.\end{aligned} \tag{23}$$

In (23), the base coordinate and the intermediate values are obtained from γ by the inverse mappings, i.e., $\mu = G^{-1}(\gamma)$, $y_Q = \mu$, $y_{\ell-1} = F_{\theta_\ell}^{-1}(y_\ell; \mathbf{c})$ for $\ell = Q, \ldots, 1$, and $\varepsilon = y_0 = T_{\boldsymbol{\theta}}^{-1}(\mu; \mathbf{c})$. Hence, ε and $y_{\ell-1}$ depend on $\boldsymbol{\theta}$ when (23) is differentiated with respect to $\boldsymbol{\theta}$. The log-PDF in (23), evaluated at the posterior signal-power particles and the corresponding contexts, supplies the terms in the learning objective $\mathcal{J}_{1:k}(\boldsymbol{\theta})$ in (21). The flow $T_{\boldsymbol{\theta}}(\,\cdot\,; \mathbf{c})$, together with the transformation $G(\cdot)$, constructs the independent field copies introduced in Section III-C. At a fixed context $\mathbf{c}$,

$$\Gamma_{\boldsymbol{\theta}}^{(k,n)}(\,\cdot\,, \mathbf{c}) = G\big(T_{\boldsymbol{\theta}}(\varepsilon_{k,n,\mathbf{c}}; \mathbf{c})\big). \tag{24}$$

The base random variables $\varepsilon_{k,n,\mathbf{c}}$ are independent across distinct copy and context indices and each is drawn from the PDF $f_0(\varepsilon_{k,n,\mathbf{c}})$. The generic variable ε in (23) denotes the base coordinate for one such draw. Hence, each field value $\Gamma_{\boldsymbol{\theta}}^{(k,n)}(\,\cdot\,,\mathbf{c})$ has the PDF $f_{\boldsymbol{\theta}}(\gamma;\mathbf{c})$ given by (23). An overview of the flow steps is shown in Fig. 3. The forward mapping in (24) draws samples of $\gamma_{k,n}$ at the context of each object, which represent the signal-power prior PDF $f_{\boldsymbol{\theta}}(\gamma_{k,n};\mathbf{c}_{k,n})$ used in Bayesian inference. The inverse mapping evaluates the log-PDF (23) at the fixed posterior signal-power particles for the flow parameter update. Before the first time step, $\boldsymbol{\theta}$ is initialized such that the flow yields the uniform PDF on $(0,\gamma_{\max}]$ for all contexts, i.e., no prior knowledge of the signal power is assumed.

V. Experiment and Evaluation

We first assess the proposed framework in a simulated multisensor radar scenario, where the signal powers are generated by a known radar-equation model (Section V-A). We then use two datasets from real-world experiments: a millimeter-wave radar dataset for meter-scale multiobject tracking in an urban environment and an acoustic dataset for kilometer-scale multiobject tracking in an underwater environment. For the acoustic setting, we additionally assess the design of the signal-power prior in a controlled simulation (Section V-C5). In all experiments, new potential objects are introduced according to the birth model in [13, Sec. II-D].

We report the generalized optimal sub-pattern assignment (GOSPA) metric [48]. This metric has three main parameters, where α controls the relative weighting of cardinality errors, p specifies the metric order, and c is the cutoff parameter for localization errors. We use the standard setting (α=2, p=1), and choose c according to the spatial scale of the localization task. At time step k, let $\mathcal{X}_k$ and $\hat{\mathcal{X}}_k$ denote the sets of true and estimated object positions, respectively. For this setting, the GOSPA metric is given by

$$\begin{aligned}\mathrm{GOSPA}(\mathcal{X}_k,\hat{\mathcal{X}}_k) = {} & \textstyle\sum_{(\mathbf{s},\hat{\mathbf{s}})\in\mathcal{D}_k}\|\mathbf{s}-\hat{\mathbf{s}}\| + \tfrac{c}{2}\big(|\mathcal{X}_k|-|\mathcal{D}_k|\big) \\ & + \tfrac{c}{2}\big(|\hat{\mathcal{X}}_k|-|\mathcal{D}_k|\big) \end{aligned} \tag{25}$$

where $\mathcal{D}_k$ denotes the assignment between true and estimated positions, i.e., a set of pairs in which each position appears at most once, that minimizes the right-hand side [48]. The three terms, i.e., the localization, missed-object, and false-object errors, correspond to the columns "Loc," "Miss," and "False" in Tables II and III. We average the metric over time steps and scenes. The GOSPA is computed based on 2-D object positions.

A. Simulated Multisensor Radar Sensing

Because the true signal-power model is not available for the real-world datasets in Sections V-B and V-C, the simulation makes it possible to compare the latent field learned during runtime directly with the power model that generated the data.

1) Simulation setup: The scenario comprises two frequency-modulated continuous-wave (FMCW) radar sensors $i\in\{1,2\}$ with low resolution, positioned at $[0\ \ 0]^{\mathrm{T}}$ and $[50\ \ 50]^{\mathrm{T}}$ in meters. Both sensors use a uniform linear array whose elements are spaced by half a wavelength and whose orientation is $\pi/4$. The measurements of each sensor constitute a separate measurement block, which has its own array manifold, signal powers $\gamma_{k,n}^{(i)}$, and noise variance $\eta_k^{(i)}$. Given these variables and the object states, the two sensors provide independent measurements. Hence, the likelihood function factorizes into two Gaussian likelihood functions of the form (3) [13], and the learning objective in (20) contains an additional sum over the sensors. For each sensor, the array manifold in (1) takes the Kronecker form

$$\mathbf{a}(\mathbf{x}) = \mathbf{a}_f\otimes\mathbf{a}_s\otimes\mathbf{a}_\beta, \tag{26}$$

$$\mathbf{v}(\xi,M) = [1,e^{j\xi},\ldots,e^{j(M-1)\xi}]^{\mathrm{T}}, \tag{27}$$

where the sensor index i is omitted and the order of the Kronecker factors corresponds to the order in which the entries of the measurement tensor are vectorized. The fast-time, slow-time, and angular factors are the Vandermonde vectors $\mathbf{a}_f=\mathbf{v}(\xi_f,M_f)$, $\mathbf{a}_s=\mathbf{v}(\xi_s,M_s)$, and $\mathbf{a}_\beta=\mathbf{v}(\xi_\beta,M_\beta)$. Their phase increments $\xi_f=4\pi\kappa T_{\mathrm{s}}R/c_0$, $\xi_s=4\pi T_{\mathrm{c}}\dot{R}/\lambda$, and $\xi_\beta=\pi\sin\beta$ depend on the range R, the radial velocity $\dot{R}$, and the angle of arrival β of the object relative to the sensor, where c_0 and λ denote the speed of light and the wavelength. The chirp rate is $\kappa=8\,\mathrm{MHz}/\mu\mathrm{s}$, the sampling interval is $T_{\mathrm{s}}=0.2\,\mu\mathrm{s}$, and the pulse repetition interval is $T_{\mathrm{c}}=40\,\mu\mathrm{s}$. Each sensor acquires $M_f=20$ samples in fast time and $M_s=10$ samples in slow time with an array of $M_\beta=10$ elements, which results in $C=2{,}000$ measurement entries.

To generate the data, each existing object is assigned a complex amplitude $\rho_{k,n}^{(i)}$ whose phase is drawn uniformly from $[0,2\pi)$ and whose squared magnitude follows the radar equation

$$\big|\rho_{k,n}^{(i)}\big|^2 = \frac{c_0^2}{f_c^2(4\pi)^3\big(R_{k,n}^{(i)}\big)^4} \tag{28}$$

with carrier frequency $f_c=77\,\mathrm{GHz}$ and range $R_{k,n}^{(i)}$ between object n and sensor i at time step k. This is the radar equation for unit transmit power, unit antenna gains, and a unit radar cross section. The amplitude magnitudes are therefore determined by the range instead of being drawn from the conditionally Gaussian model in (2). We set the noise variance such that an object located at the center of the region of interest (ROI) $[0,50]\times[0,50]\,\mathrm{m}$ has a signal-to-noise ratio per measurement entry of $10\log_{10}\big(|\rho_{k,n}^{(i)}|^2/\eta_k^{(i)}\big)=-3\,\mathrm{dB}$.

Over 80 time steps, five objects move inside the ROI following a constant-velocity model [49, Sec. 6.3.2], and their kinematic states are composed of 2-D position and velocity. The initial states of objects 1 and 2 are $[10\ \ 25\ \ 0.5\ \ 0]^{\mathrm{T}}$ and $[25\ \ 10\ \ 0\ \ 0.5]^{\mathrm{T}}$, where the position precedes the velocity, and these objects are present from time step 2 to 72 and from 5 to 75, respectively. Objects 3, 4, and 5 start from random initial states and are present from time step 10 to 70, 18 to 78, and 25 to 75, respectively. Both sensors share a single latent

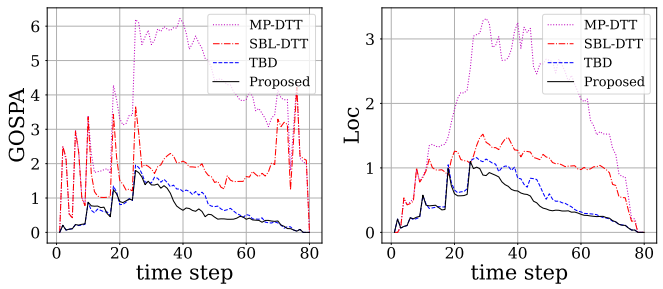


Fig. 4. Total GOSPA error (left) and localization error (right) as functions of time in the simulated multisensor radar scenario, averaged over 100 runs. In the legend, "MP-DTT," "SBL-DTT," and "TBD" denote MP + BP, SBL + BP, and BP-TkBD [13], respectively. Both errors are given in meters.

field, i.e., a single flow with parameters $\boldsymbol{\theta}$, whose context $\mathbf{c}_{k,n}^{(i)}$ consists of the estimated range and angle of the object with respect to sensor i. All results are averaged across 100 Monte Carlo runs.

2) Baselines: Three reference methods are considered. MP + BP and SBL + BP are two detect-then-track methods. Point detections are extracted by matching pursuit (MP) [50] or SBL [21], [51], respectively, and then processed by the multisensor BP tracking method in [52] with 30,000 kinematic-state particles. BP-TkBD [13] operates on the same raw measurements as the proposed method, but its signal powers evolve according to a prescribed random-walk model. For both BP-TkBD and the proposed method, the kinematic states, signal powers, and noise variances are represented by 10,000, 500, and 500 particles, respectively. The GOSPA cutoff is set to $c = 5\,\mathrm{m}$.

3) Results: Fig. 4 depicts the time evolution of the GOSPA error and its localization component. The largest errors are obtained by the detect-then-track baselines. For SBL + BP, the error exhibits peaks at the time steps where objects appear or disappear, which is in line with missed detections, false detections, and data-association errors that occur at the detection stage. Considerably lower errors are achieved by the two TkBD methods. For approximately the first 30 time steps, during which the flow has been updated with only limited posterior information, the proposed method and BP-TkBD perform similarly. Afterward, the proposed method yields lower GOSPA and localization errors than BP-TkBD, and the gap is largest between approximately $k = 35$ and $k = 55$. At the end of the sequence, the errors of the two methods are again similar. This suggests that the latent field describes the range-dependent signal powers more accurately than the random-walk model.

For visualization, we use the mean field, i.e., the mean signal power of the latent field at context $\mathbf{c}$, given by

$$m_{\boldsymbol{\theta}}(\mathbf{c}) = \mathbb{E}\big[\Gamma_{\boldsymbol{\theta}}(\cdot\,, \mathbf{c})\big] = \int_0^\infty \gamma\, f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})\, \mathrm{d}\gamma \qquad (29)$$

which we approximate by the sample mean of samples drawn from $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c})$. For sensor 1, the mean field is evaluated at the cells of a regular grid with a cell size of $0.5 \times 0.5\,\mathrm{m}^2$, using 512 samples per cell. Fig. 5 juxtaposes this map with the power predicted by the radar equation (28). Over the ROI, both maps decrease by about 50 dB from the region near the sensor to the far corner, while their pointwise difference stays below about 2 dB. The latent field, which is learned during runtime and without labeled data, thus accurately captures the range dependence of the signal power used to generate the data.

Tracking results at selected time steps, which show that the proposed method resolves closely spaced objects despite the low sensor resolution, are provided in [44, Sec. IV].

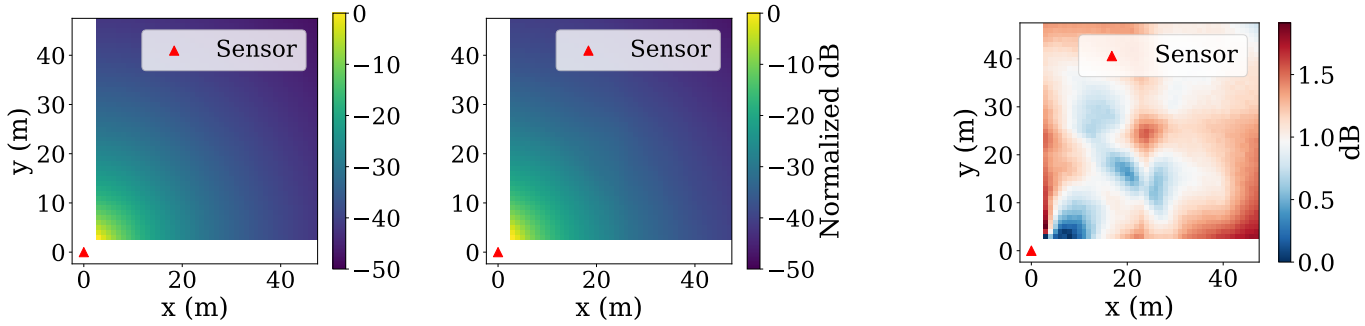
B. Active Radio Sensing

We use the UWCR dataset [53], which provides raw analog-to-digital converter (ADC) data from a 77 GHz FMCW signal. A Texas Instruments AWR1843 board equipped with 2 transmit and 4 receive antennas serves as a radar sensor. At each time step, the ADC data of one radar frame is a four-dimensional complex-valued tensor of size $128 \times 255 \times 2 \times 4$ (fast-time samples $\times$ chirps $\times$ transmitters $\times$ receivers), yielding a measurement dimension of $C = 261{,}120$.

1) Implementation details: We first concatenate the two transmit paths and four receive paths into eight virtual channels and reorder each frame as a $128 \times 8 \times 255$ tensor (fast time $\times$ virtual array $\times$ slow time). The proposed method consumes this complex-valued tensor directly: apart from scaling by a fixed normalization constant for numerical stability, it does not construct an intermediate range–Doppler (RD) or RA intensity map. A 3-D FFT is used only to initialize new potential objects: it identifies coarse search regions from peak responses without CFAR. A Gaussian birth PDF $f_{\mathrm{B},n}(\mathbf{x}_{k,n})$ is centered at the peak in each search region and is subsequently refined by sequential Bayesian inference on the raw ADC samples. Thus, the FFT supplies only an initial search region and does not provide the measurements used in the subsequent state updates.

For this FMCW configuration, we instantiate the physics-based array manifold in (1) using the Kronecker structure in (26), with the Kronecker factors ordered according to the vectorization of the $128 \times 8 \times 255$ tensor, where the fast-time, slow-time, and virtual-array factors all have the Vandermonde form in (27), with phase increments determined by range, radial velocity, and azimuth, respectively. These phase increments are given by the expressions in Section V-A, evaluated with the chirp parameters of the dataset; as in Section V-A, the virtual array has half-wavelength element spacing, and β denotes the azimuth. This factorization retains the phase response prescribed by the radar geometry while the latent field models the context-dependent signal-power PDFs. The unit-magnitude Vandermonde factors give $\|\mathbf{a}(\mathbf{x})\|_2^2 = C$, so an existing object's expected squared signal norm is $C\gamma_{k,n}$ in the measurement scaling used for inference.

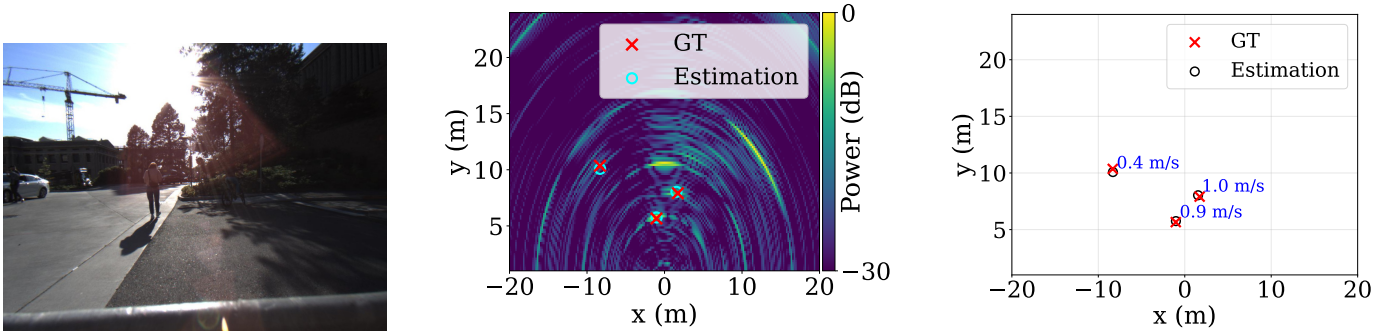
The normalizing-flow context contains positional encodings of the range and azimuth obtained by MMSE estimation from the previous posterior, together with estimated radar-cross-section (RCS) statistics of each object. The flow is initialized at the beginning of each sequence and trained during runtime, without a separate training set or offline pretraining. We use Adam with a learning rate of 10^{-4} and perform five gradient steps per time step. Forward Bayesian inference uses 10,000



(a) Radar equation (left) and mean field (right)

(b) Difference between the maps in (a)

Fig. 5. Signal power of sensor 1 in the simulated multisensor radar scenario. (a) Normalized power in dB according to the radar equation (28) (left) and according to the mean field $m_{\boldsymbol{\theta}}(\mathbf{c})$ in (29) (right). (b) Pointwise deviation of the learned map from the radar-equation map.

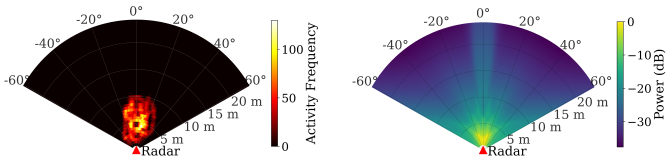


(a) Camera reference

(b) Observed power

(c) Est. positions and velocities

Fig. 6. Example scene from the UWCR dataset. (a) Synchronized camera view in an urban environment. (b) Observed spatial power computed from the received measurement. (c) Estimated positions and velocities of all moving objects. In (b) and (c), "GT" denotes the ground truth.



(a) Activity map

(b) Mean field of the learned latent field

Fig. 7. Visualization of the activity map and the learned latent field in a parking-lot environment.

particles for the object state $\mathbf{x}$, 3,000 particles for the signal power γ, and 500 particles for the noise variance η. Additional details about the dataset are provided in [53].

2) Baselines: We use the GOSPA metric with a cutoff of $c = 5\,\text{m}$, averaged across all scenes and 50 Monte Carlo runs. The methods in Table II comprise detector-only methods, detect-then-track methods, and the proposed raw-measurement method. The classical detectors apply moving target indication (MTI) [54], windowed range and Doppler FFTs, non-coherent accumulation, 2-D CFAR, angle estimation, and density-based spatial clustering of applications with noise (DBSCAN) [55]. They differ in the angle estimator, which is a 3-D FFT that forms a range–angle–Doppler (RAD) cube, MUSIC [19], or the minimum variance distortionless response (MVDR) beamformer [5]. MUSIC and MVDR use spatial smoothing and are applied only to the array snapshots at CFAR-selected RD cells, which yields the representation "RD Map + CFAR Dets." in Table II. The learning-based detectors RODNet [7], which we retrain on the UWCR dataset, and RAMP-CNN [6], for which we use the authors' checkpoint, process FFT-derived RA maps and, in the case of RAMP-CNN, additional RV and VA maps; for both, we retain only the positions of moving objects and obtain point detections by clustering. The tracking methods BP [29], the probability hypothesis density (PHD) filter [28], and the Poisson multi-Bernoulli mixture (PMBM) filter [27] process these point detections, whereas the proposed method updates the multiobject posterior directly from the raw complex ADC samples after coarse birth initialization. Since the methods also differ in their measurement models and priors, this comparison does not isolate the effect of avoiding intermediate measurement compression.

3) Results: Table II compares the proposed method with different detectors and tracking methods based on the UWCR dataset. The detectors rely on compressed representations, and the tracking methods operate on the resulting detection sets. These representations provide efficient summaries of the measurements but may discard information useful for downstream localization. Our method directly takes the complex ADC data as input without generating a compressed representation in a preprocessing step, apart from the coarse birth initialization. Table II indicates that our method achieves the lowest GOSPA. These results indicate that combining inference based on the raw data with a learned latent field is beneficial for this multiobject tracking task.

Table II also reports the average runtime per frame on an NVIDIA RTX 4090 with single-precision arithmetic; for

TABLE II
PERFORMANCE COMPARISON BASED ON THE UWCR DATASET. ↓ INDICATES LOWER IS BETTER. GOSPA VALUES ARE GIVEN IN METERS, RUNTIMES ARE GIVEN IN MILLISECONDS PER FRAME, AND BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN. FOR DETECT-THEN-TRACK METHODS, THE RUNTIMES OF THE DETECTOR AND THE TRACKING METHOD ARE LISTED SEPARATELY. PERFORMANCE IS AVERAGED OVER 50 MONTE CARLO RUNS. WE APPLIED THE PROPOSED METHOD WITH AND WITHOUT THE LATENT FIELD.

Method	Methodology	Key Representation	GOSPA ↓	Loc ↓	Miss ↓	False ↓	Runtime ↓
3-D FFT	Detector	RAD Cube	1.187	0.726	0.342	0.119	**3.15**
MUSIC [19]	Detector	RD Map + CFAR Dets.	1.142	0.722	0.312	0.107	4.81
MVDR [5]	Detector	RD Map + CFAR Dets.	1.125	0.707	0.326	0.093	4.57
RODNet [7]	Detector	Complex RA maps	1.077	0.698	0.100	0.279	420.28
RAMP-CNN [6]	Detector	RA/RV/VA maps	0.802	0.708	0.050	0.044	414.25
MVDR + BP [29]	Det. + Track	Point Cloud	0.871	0.759	0.048	0.065	4.57 + 8.86
MVDR + PMBM [27]	Det. + Track	Point Cloud	0.877	0.753	0.089	0.035	4.57 + 8.63
MVDR + PHD [28]	Det. + Track	Point Cloud	1.105	0.696	0.366	0.044	4.57 + 0.27
RAMP-CNN + BP [29]	Det. + Track	Point Cloud	0.765	0.690	0.040	0.035	414.25 + 9.18
Proposed w/o latent field	TkBD	Raw Complex (ADC)	0.577	0.542	0.017	0.018	29.09
Proposed	TkBD	Raw Complex (ADC)	**0.552**	**0.526**	**0.013**	**0.013**	69.86

the learning-based detectors, it comprises network inference but excludes offline training and separately executed preprocessing. All runtimes are measured on the same workstation, which is equipped with an AMD Ryzen Threadripper PRO 5955WX CPU with 16 cores (32 threads), 128 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory, using Python 3.8 with PyTorch 2.0.1 and CUDA 11.8, running on Ubuntu 22.04.The proposed method requires $69.86\,\text{ms}$ per frame, of which $21.45\,\text{ms}$, $45.06\,\text{ms}$, and $3.34\,\text{ms}$ are spent on forward Bayesian inference, the flow update, and prior generation and initialization, respectively. This runtime exceeds those of the classical detectors and of the MVDR-based detect-then-track methods but is substantially lower than those of the learning-based detectors. Thus, the low-rank and Kronecker-structured likelihood evaluation makes sequential inference based on approximately 2.6×10^5 complex ADC samples per frame computationally feasible.

Fig. 6 shows a synchronized camera view of an example scene, the observed spatial power computed by beamforming, and the positions and velocities of the moving objects estimated by the proposed method. Although the strongest responses in the observed power are not located at the object positions, the estimated positions agree closely with the ground truth. A particle representation of the posterior PDF, which is sharply concentrated because the likelihood function of the proposed signal model is highly informative, is provided in [44, Sec. V].

Fig. 7 visualizes the latent field learned during runtime by the normalizing flow in a parking-lot environment. The activity map in Fig. 7(a) aggregates the spatial regions occupied by localized objects over all time steps, and therefore indicates the area from which measurements are collected. Fig. 7(b) shows the mean field (29) evaluated on a dense spatial grid. The learned field exhibits a dominant mainlobe around 0° and weaker sidelobes near $\pm 20^\circ$ and $\pm 60^\circ$, which may reflect the radiation pattern of the radar antennas, and the power decreases with increasing range. Hence, the learned structure is not merely a copy of the activity map: while the activity map reflects where objects have been observed, the learned latent field describes how the mean signal power varies across space. In regions that are not covered by the activity map, e.g., at ranges below $5\,\text{m}$ and near $\pm 60^\circ$, the latent field is an extrapolation determined by the context parameterization. For this visualization, we intentionally omit object-specific features and use the polynomial positional context

$$\mathbf{c} = \left[\bar{R}, \bar{R}^2, \ldots, \bar{R}^U, \{\sin^2(u\bar{\beta}), \cos(u\bar{\beta})\}_{u=1}^{U}\right]^{\mathrm{T}}, \qquad (30)$$

where $\bar{R}$ and $\bar{\beta}$ denote normalized range and azimuth, respectively, and U is the order of the encoding. This context encourages the flow to represent spatial propagation effects and enforces symmetry about 0°; hence, the symmetry in Fig. 7(b) results from this choice of context and is not learned from the data.

The evolution of the signal-power PDF through the successive flow transformations is illustrated in [44, Sec. V].

Replacing the 3-D FFT peak that serves as the mean of each Gaussian birth PDF by detections from MVDR or RAMP-CNN changes the GOSPA by less than $0.005\,\text{m}$ [44, Sec. V]. Hence, the birth PDF needs only to identify a plausible search region.

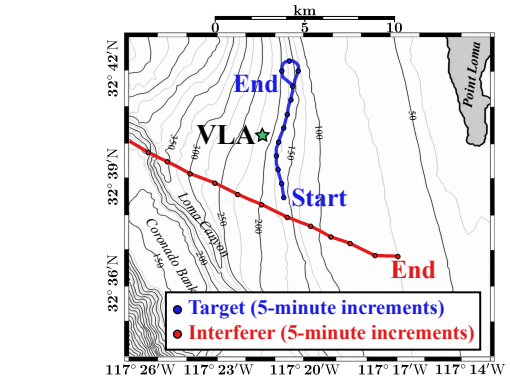
C. Passive Acoustic Sensing

For this underwater acoustic multiobject tracking problem, we use the SWellEx-96 dataset [57]. In particular, we use data from Events S5 and S59 recorded on May 11, 1996 between 00:05–00:25 and on May 13, 1996 between 12:00-12:20, respectively. The experiment was conducted approximately $12\,\text{km}$ southwest of Point Loma, San Diego, California, in water approximately $216.5\,\text{m}$ deep. A 21-element vertical line array (VLA) serves as the receiver. The array spans an aperture of $120\,\text{m}$. Fig. 8 shows an example layout and corresponding estimation results, where the R/V Sproul towed an underwater acoustic source (blue) northward along an isobath while a surface-ship interferer (red) simultaneously traversed the area.We use seven tonal signals transmitted by the underwater source, with frequencies from 49 Hz to 388 Hz, while the receiver records acoustic pressure at the location of the 21 elements.

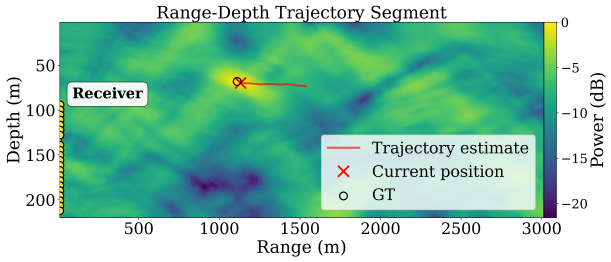
1) Implementation details: The physical measurements are the raw time series from the 21 VLA hydrophones. We apply the same front end to all methods: short-time Fourier

TABLE III
PERFORMANCE COMPARISON BASED ON THE SWELLEX-96 DATASET. ↓ INDICATES LOWER IS BETTER. GOSPA VALUES ARE GIVEN IN METERS, AND BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN.

Method	Methodology	Key Representation	GOSPA ↓	Loc ↓	Miss ↓	False ↓
MP [50]	Detector	Raw Complex	35.973	30.322	**2.825**	2.825
SBL [21]	Detector	Raw Complex	36.312	30.148	3.253	2.911
MP + BP [29]	Det. + Track	Point Cloud	32.377	29.123	3.082	0.172
SBL + BP [29]	Det. + Track	Point Cloud	32.624	29.285	3.082	0.257
Proposed	TkBD	Raw Complex	**30.832 ± 0.09**	**27.835**	2.997	**0.000**



(a) Sensing environment



(b) Estimated trajectory over 5 minutes

Fig. 8. Event S59 of the SWellEx-96 dataset. (a) Sensing environment. The blue trajectory corresponds to the source to be tracked, which is labeled "Target" in the legend, and the red trajectory corresponds to an interfering source. (b) Trajectory estimated by the proposed method. The background depicts the spatial power of the acoustic pressure field obtained by matched-field processing [56].

analysis extracts complex coefficients at the seven transmitted tones, and the measurement of each time step consists of three consecutive snapshots. The hydrophone signals are sampled at 1500 Hz, and the short-time Fourier analysis uses a Hamming window with an FFT length of 8,192 samples, i.e., approximately 5.46 s, and an overlap of 75%. Consecutive snapshots are therefore 2,048 samples apart, and the time between consecutive time steps is 4.096 s. Each tone defines a dictionary, and each tone/snapshot block contains 21 complex hydrophone coefficients. We use the observation model for multiple dictionaries and snapshots in [13, Sec. II-A]: kinematic states and existence variables are shared across blocks, signal powers and noise variances are dictionary-specific and shared across snapshots, and block likelihoods multiply conditioned on these states. The proposed method performs inference jointly across these tones and snapshots without first reducing them to point detections.

For an underwater source state $\mathbf{x} = [R, \dot{R}, z, \dot{z}]^{\mathrm{T}}$, where R, $\dot{R}$, z, and $\dot{z}$ denote range, range rate, depth, and depth rate, respectively, as in [13], the array manifold depends only on R and z and must account for multipath propagation and dispersion in the shallow-water waveguide rather than using a far-field plane-wave approximation. For one tone, with its frequency index suppressed, we define the 21-component unit-norm replica vector $\mathbf{a}(\mathbf{x}) = \mathbf{p}(\mathbf{x})/\|\mathbf{p}(\mathbf{x})\|_2 \in \mathbb{C}^{21}$, where $\mathbf{p}(\mathbf{x})$ is the complex pressure vector with entries $p_i(R,z) \approx \sum_{m=1}^{M} \Psi_m(z)\Psi_m(z_i)\frac{e^{jk_m R}}{\sqrt{k_m R}}$ for each hydrophone $i \in \{1,\dots,21\}$. Here, $\Psi_m(\cdot)$ is the depth function of mode m, k_m is its horizontal wavenumber, and z_i is the depth of hydrophone i [58]. The unit-norm replica preserves the relative complex response across hydrophones and satisfies $\|\mathbf{a}(\mathbf{x})\|_2^2 = 1$. Thus, an existing object's coefficient power equals its expected squared signal norm within one tone/snapshot block, in the chosen measurement scaling. This formulation follows the matched-field processing paradigm [56], in which the received array data are coherently compared with modeled wavefields for candidate source positions. Note that the interferer is not tracked, since its contribution to the measurements is suppressed by the matched-field processing approach that only considers field values at the seven tones transmitted by the acoustic source [56].

The flow context contains the tone frequency and a positional encoding of the MMSE range and depth estimates from the preceding posterior, which corresponds to the flow with spatial context only in the ablation study of Section V-C5. For example, the normalized range $\bar{R}$ is encoded as

$$[\sin(2^0\pi\bar{R}), \cos(2^0\pi\bar{R}), \dots, \sin(2^{U-1}\pi\bar{R}), \cos(2^{U-1}\pi\bar{R})]^{\mathrm{T}}$$

with an analogous encoding for depth, where U is the number of encoding frequencies. We train the normalizing flow during runtime using Adam with a learning rate of 10^{-4} and perform five gradient steps per time step. The source follows a constant-velocity motion model in range and depth with zero-mean Gaussian driving noise of covariance $\mathrm{diag}(10^{-3}, 10^{-5})$, as described in [13]. Sequential inference uses 10,000 particles for $\mathbf{x}$, 3,000 particles for γ, and 1,000 particles for η.

2) Baselines: We use GOSPA with a cutoff $c = 100$ m for evaluation. MP [50] represents the measurements of each time step as a sparse combination of range–depth replica vectors. It greedily selects the dictionary atom most aligned with the residual energy averaged across tones and snapshots, estimates its coefficient, and removes its contribution. SBL [21] uses the same dictionary but estimates a sparse probabilistic representation by evidence maximization, shrinking the variances of inactive grid cells toward zero. The detect-then-track variants pass the resulting point detections to BP [29]. In contrast, the proposed method retains the joint complex measurements

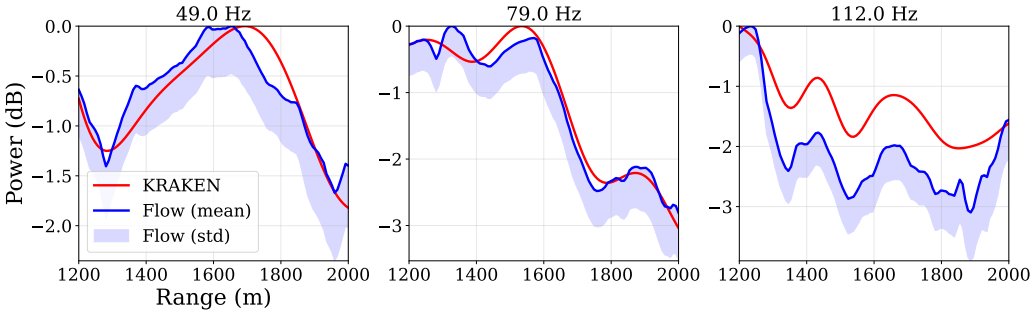


Fig. 9. Normalized power as a function of range at a source depth of 70 m for the three lowest tones at 49, 79, and 112 Hz, according to KRAKEN and to the learned latent field. Each curve is normalized to a maximum of 0 dB. For the latent field, the mean field $m_{\boldsymbol{\theta}}(\mathbf{c})$ and, for readability, only the lower half of the ± 1 standard-deviation band (shaded) are shown.

across all seven tones and three snapshots and conditions its normalizing flow on the estimated source location and tone frequency.

This comparison therefore parallels the radio experiment: the baseline detectors and tracking methods operate on an intermediate sparse representation, whereas the proposed method performs the posterior update before such compression. The acoustic setting additionally tests whether the learned latent field can recover frequency-dependent propagation structure in a dispersive multipath environment.

3) Results: Table III leads to similar conclusions as in the radio experiment in Section V-B. Detector-only methods exhibit the largest errors. Adding a tracking method on top of the detector improves performance by exploiting structure in time and space, but the overall accuracy remains limited by the quality of the upstream point estimates. In contrast, the proposed method achieves the lowest GOSPA as well as the lowest localization and false-object errors by jointly exploiting raw complex measurements and a learned latent field.

4) Comparison with KRAKEN: We compare the learned latent field with the power predicted by the KRAKEN normal-mode program [59]. Fig. 9 shows the normalized power of both fields as a function of range for the three lowest tones at 49, 79, and 112 Hz. KRAKEN computes a deterministic propagation prediction from an idealized waveguide model specified by the sound-speed profile, seabed parameters, and array geometry. We use this prediction as a physics-based reference rather than ground truth: the flow is trained only through the variational objective and is not supervised by KRAKEN amplitudes.

For a quantitative comparison, we evaluate the mean field $m_{\boldsymbol{\theta}}(\mathbf{c})$ and the KRAKEN-predicted power as a function of range for the fixed estimated source depth of 70 m. Note that in this mildly range-dependent environment, the estimated source position is deeper than the true source position because of the mirage effect [60]. It can be seen that the learned and KRAKEN-predicted fields follow the same range-dependent trend at low frequencies. The corresponding curves for the tones at 148, 201, 283, and 388 Hz, which are provided in [44, Sec. VI], show a growing gap at higher frequencies. At these frequencies and ranges, volume absorption in seawater is negligible, whereas bottom attenuation and boundary scattering increase with frequency and can be underrepresented by an idealized normal-mode calculation. Moreover, the modes at higher frequencies are more sensitive to small mismatches in the sound-speed profile, seabed properties, and array geometry. Consequently, KRAKEN itself may also deviate from the unknown true range dependence of the signal power at these frequencies. In addition, the latent field may be less accurate at higher frequencies, because the modal interference pattern varies on shorter range scales, which the chosen positional encoding may not resolve.

5) Ablation study of the signal-power prior: In a controlled acoustic simulation in which KRAKEN generates the measurements, we compare flow-based and parametric signal-power priors while keeping the measurement model, inference procedure, and object trajectories fixed. A flow whose context contains summary statistics of the preceding posterior achieves the lowest GOSPA of 21.44, compared with 22.13 for a flow with spatial context only, 23.57 for a flow with a shifted base PDF, 27.16 for a context-conditioned Gamma prior, and 31.92 for a Gamma random walk. These results indicate that both spatial conditioning and a flexible density model are important, and that temporal information is most effective when it is supplied through the flow context. The simulation setup, the compared priors, and a detailed discussion are provided in [44, Sec. VII].

VI. Conclusion

We proposed a Bayesian framework for multiobject tracking that directly exploits raw complex measurements. Our approach integrates a physics-based array model with a learnable latent field that is parameterized by a normalizing flow and optimized during runtime via variational inference. Simulations with a known radar-equation power model showed that the latent field learned during runtime reproduces this model closely. Evaluations on both active radio and passive acoustic datasets confirm that our framework outperforms conventional baselines. We demonstrate that performing inference on raw complex-valued signals and learning a latent field that represents objects and the propagation environment can offer a scalable new paradigm for wireless sensing. The new paradigm can lead to a safer navigation of robots, improved situational awareness in search and rescue, environmental monitoring, and related applications.

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